

①
25.05.2023 (E)



SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: January 2023 – May 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration: 3 Hrs.
Programme code: 01		Semester: VI (SVU 2020)
Programme: B.Tech Computer Engineering		Class: TY
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116U01E622	Name of the Course: Data Mining and Business Intelligence	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks												
Q1	Solve any TWO	20												
i)	Explain Multilevel Association Rules and Multidimensional Association Rules with relevant examples.	10												
ii)	Differentiate between simple linkage, average linkage and complete linkage algorithms. Use complete linkage algorithm to find the clusters from the following dataset. <table><tr><td>X</td><td>4</td><td>8</td><td>15</td><td>24</td><td>24</td></tr><tr><td>Y</td><td>4</td><td>4</td><td>8</td><td>4</td><td>12</td></tr></table>	X	4	8	15	24	24	Y	4	4	8	4	12	10
X	4	8	15	24	24									
Y	4	4	8	4	12									
iii)	Describe various methods for handling missing values in real-world data.	10												
Q2 A	Perform k-medoids for the following data points with k=2 (8,7), (3,7), (4,9), (9, 6), (8, 5), (5,8)	10												
	OR													
Q2 A	A node in the decision tree represents a splitting attribute and the branch of the node represents the different outcomes i.e. different values the node can take. 1. Explain the different types of partitioning the tuples when the splitting attribute is a) discrete-valued b) discrete-valued with only 2 outcomes c) continuous valued. 2. Exemplify your answers	10												
Q2 B	Solve any One	10												
i)	Explain in detail Click stream mining.	10												
ii)	The following table consists of training data from an employee database. The data have been generalized. For example, "31 ... 35" for age represents the age range of 31 to 35. For a given row entry, ... represents the number of data tuples	10												

having the values for *department*, *status*, *age*, and *salary* given in that row.

<i>department</i>	<i>status</i>	<i>age</i>	<i>salary</i>
sales	senior	31 ... 35	46K ... 50K
sales	junior	26 ... 30	26K ... 30K
sales	junior	31 ... 35	31K ... 35K
systems	junior	21 ... 25	46K ... 50K
systems	senior	31 ... 35	66K ... 70K
systems	junior	26 ... 30	46K ... 50K
systems	senior	41 ... 45	66K ... 70K
marketing	senior	36 ... 40	46K ... 50K
marketing	junior	31 ... 35	41K ... 45K
secretary	senior	46 ... 50	36K ... 40K
secretary	junior	26 ... 30	26K ... 30K

Let *status* be the class label attribute.

(i) Use your algorithm to construct a decision tree from the given data.

(ii) Given a data tuple having the values "*systems*," "*26 ... 30*," and "*46-50K*" for the attributes *department*, *age*, and *salary*, respectively, what would a naive Bayesian classification of the *status* for the tuple be?

Q3	Solve any Two	20
i)	Briefly outline with example, how to compute the dissimilarity between objects described by the following: i. Nominal Attributes ii. Asymmetric binary attributes	10
ii)	Suppose that the data mining task is to cluster the following eight points (with (x, y) representing location) into three clusters. A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9). The distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. Use the k-means algorithm to show only (i) The three cluster centers after the first round of execution and (ii) The final three clusters	10
iii)	Explain why data integration is required as a part of pre-processing in data mining? List all the methods of data reduction. Explain in detail the stratified sampling technique with an example applying it for any real time data.	10
Q4	Solve any Two	20
i)	Discuss examples of any 5 data mining tasks using real world database.	10
ii)	Explain KDD Process with a relevant diagram. Discuss the applications and issues in data mining.	10
iii)	A database has five transactions. Let min sup = 60% and min conf = 80%. (a) Find all frequent itemsets using Apriori (b) Generate association rules	10

		Transaction – ID	Items		
		t100	M, O, N, K, E, Y		
		t200	D, O, N, K, E, Y		
		t300	M, A, K, E		
		t400	M, U, C, K, Y		
		t500	C, O, O, K, I, E		
Q5	Write short notes on any four				20
i)	Business Intelligence Systems				5
ii)	Lift				5
iii)	Correlation Analysis				5
iv)	Logistic Regression.				5
v)	Decision Support System				5



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Semester: January 2023 –May 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01		
Programme: B. Tech. (Computer Engineering)	Class: TY	Semester: VI(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116U01E626	Name of the Course: Machine Learning	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

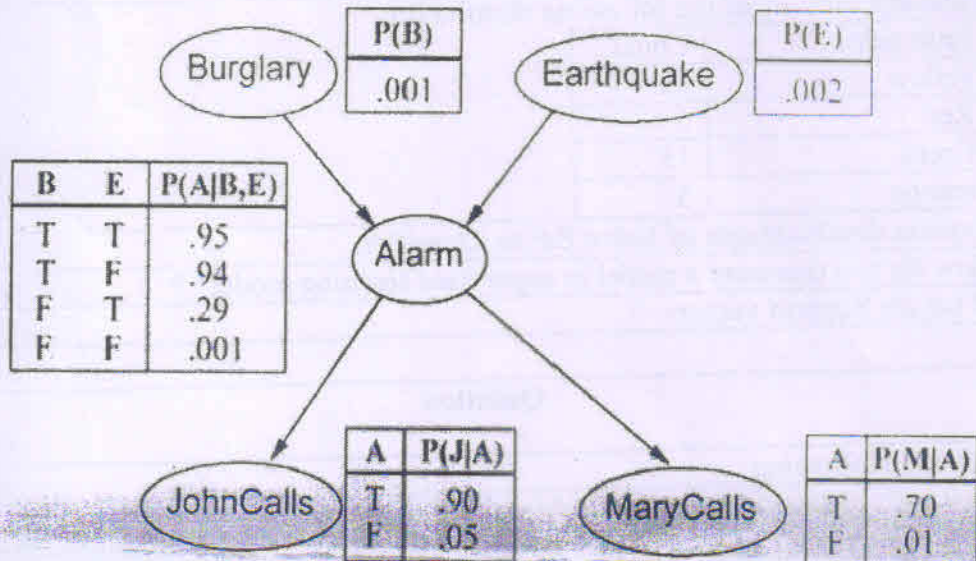
Que. No.	Question	Max. Marks										
Q1	Solve any Four	20										
i)	Define Machine learning and how are experience, performance and tasks related with Machine learning.	5										
ii)	Discuss application of Logistic Regression.	5										
iii)	Calculate entropy of the following distribution <table><tr><td>Fruit color</td><td>Count</td></tr><tr><td>yellow</td><td>10</td></tr><tr><td>Red</td><td>5</td></tr><tr><td>Green</td><td>15</td></tr><tr><td>orange</td><td>5</td></tr></table>	Fruit color	Count	yellow	10	Red	5	Green	15	orange	5	5
Fruit color	Count											
yellow	10											
Red	5											
Green	15											
orange	5											
iv)	Explain disadvantages of Naïve Bayes Classifier.	5										
v)	How do you represent a model in supervised learning model ?	5										
vi)	What are Support vectors ?	5										

Que. No.	Question	Max. Marks																
Q2 A	Solve the following	10																
i)	What is Expectation Maximization Algorithm ?	5																
ii)	Explain Random Forest ensemble technique for Machine learning.	5																
	OR																	
Q2 A	With block diagram explain how Machine learning works.	10																
Q 2 B	Solve any One																	
i)	For the following confusion matrix calculate Accuracy, True Positive Rate, False Positive Rate and F1 score.	10																
	<table><tr><td colspan="2">Confusion Matrix for Apples and Bananas</td><td colspan="2">Actual class</td></tr><tr><td></td><td></td><td>Banana</td><td>Apple</td></tr><tr><td>Predicted class</td><td>Banana</td><td>5</td><td>2</td></tr><tr><td></td><td>Apple</td><td>3</td><td>3</td></tr></table>	Confusion Matrix for Apples and Bananas		Actual class				Banana	Apple	Predicted class	Banana	5	2		Apple	3	3	
Confusion Matrix for Apples and Bananas		Actual class																
		Banana	Apple															
Predicted class	Banana	5	2															
	Apple	3	3															

ii) Given the following data for BMI and cholesterol for 10 patients. Predict the likely value of cholesterol for someone who has BMI of 27. Use linear regression.

10

BMI	Cholesterol
17	140
21	189
24	210
28	240
14	130
16	100
19	135
22	166
15	130
18	170

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain in brief the working of the Support Vector Machine.	10
ii)	Explain Hidden Markow Model with example.	10
iii)	 a) What is the probability that the alarm has sounded but neither a burglar nor earthquake has occurred and both John and Merry call? a) What is the probability that John call ? Use Bayesian Belief Networks for the above problem.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Apply K-means algorithm in given data for $K = 3$, clusters are 3. Use $C1(2)$, $C2(16)$ and $C3(38)$ as initial cluster centers. Data: 2,4,6,3,31,12,15,16,38,35,14,21,23,25,30	10

ii)

Sepal Length	Sepal Width	Species
5.3	3.7	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor

10

Use K nearest Neighbour technique to find

Sepal Length	Sepal Width	Species
5.2	3.1	?

iii)

Solve using perceptron learning rule to train the network where c is learning constant = 0.1, desired response for X_1 , X_2 and X_3 are $d_1 = -1$, $d_2 = -1$, $d_3 = 1$. All inputs are column vectors.

$$X_1 = [1, -2, 0, 1]$$

$$X_2 = [0, 1.5, -0.5, -1]$$

$$X_3 = [-1, 1, 0.5, -1]$$

$$W_1 = [1, -1, 0, 0.5]$$

Calculate weight after one complete cycle.

10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	What is goal of Support Vector Machine	5
ii)	Give one application of Bayesian Belief Networks.	5
iii)	What are elements of reinforcement learning	5
iv)	List one application of Neural Network.	5
v)	What is the need for Dimensionality Reduction	5
vi)	What are types of Machine Learning	5



Semester: January 2023 –May 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: : 41		
Programme: Minor Programme in Artificial Intelligence and Machine Learning	Class: TY	Semester: VI
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116m41C601	Name of the Course: Machine Learning (Minor)	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What is the basic principle of support vector machines?	5
ii)	How attribute selection is performed in Decision tree learning algorithm ?	5
iii)	How logistic regression function is used in classification?	5
iv)	Why K-nearest neighbor is called as a lazy learning algorithm?	5
v)	What is overfitting and underfitting of data in decision tree classifier? How to avoid it?	5
vi)	Write the steps to carry out PCA. Explain with mathematical notations.	5

Que. No.	Question	Max. Marks																						
Q2 A	Solve the following	10																						
i)	What is the role of activation function in ANN?	5																						
ii)	Why multilayer perceptron is more useful than a single layer perceptron?	5																						
	OR																							
Q2 A	A laboratory have following data of BMI and cholesterol level for 10 patients. Predict the likely value of cholesterol level who have BMI value as 27. Justify the answer.	10																						
	<table><tr><th>BMI</th><th>Cholesterol</th></tr><tr><td>17</td><td>140</td></tr><tr><td>21</td><td>189</td></tr><tr><td>24</td><td>210</td></tr><tr><td>28</td><td>240</td></tr><tr><td>14</td><td>130</td></tr><tr><td>16</td><td>100</td></tr><tr><td>19</td><td>135</td></tr><tr><td>22</td><td>166</td></tr><tr><td>15</td><td>130</td></tr><tr><td>18</td><td>170</td></tr></table>	BMI	Cholesterol	17	140	21	189	24	210	28	240	14	130	16	100	19	135	22	166	15	130	18	170	
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Q 2 B	Solve any One	10																																																													
i)	Consider the following data. Apply KNN classifier . If John has height 161 and weight 61, which is the correct T-shirt size for John?	10																																																													
<table><tr><th>height</th><th>weight</th><th>T-Shirt size</th></tr><tr><td>158</td><td>58</td><td>M</td></tr><tr><td>158</td><td>59</td><td>M</td></tr><tr><td>158</td><td>63</td><td>M</td></tr><tr><td>160</td><td>59</td><td>M</td></tr><tr><td>160</td><td>60</td><td>M</td></tr><tr><td>163</td><td>60</td><td>M</td></tr><tr><td>163</td><td>61</td><td>M</td></tr><tr><td>160</td><td>64</td><td>L</td></tr><tr><td>163</td><td>64</td><td>L</td></tr><tr><td>165</td><td>61</td><td>L</td></tr><tr><td>165</td><td>62</td><td>L</td></tr><tr><td>165</td><td>65</td><td>L</td></tr><tr><td>168</td><td>62</td><td>L</td></tr><tr><td>168</td><td>63</td><td>L</td></tr><tr><td>168</td><td>66</td><td>L</td></tr><tr><td>170</td><td>63</td><td>L</td></tr><tr><td>170</td><td>64</td><td>L</td></tr><tr><td>170</td><td>68</td><td>L</td></tr></table>			height	weight	T-Shirt size	158	58	M	158	59	M	158	63	M	160	59	M	160	60	M	163	60	M	163	61	M	160	64	L	163	64	L	165	61	L	165	62	L	165	65	L	168	62	L	168	63	L	168	66	L	170	63	L	170	64	L	170	68	L				
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170	64	L																																																													
170	68	L																																																													
ii)	Consider the following table for binary classification. Calculate the root of the decision tree using Gini Index.	10																																																													
<table><tr><th>Customer income</th><th>Gender</th><th>Car type</th><th>class</th></tr><tr><td>High</td><td>M</td><td>Family</td><td>C1</td></tr><tr><td>High</td><td>M</td><td>Sports</td><td>C1</td></tr><tr><td>High</td><td>M</td><td>family</td><td>C2</td></tr><tr><td>low</td><td>M</td><td>family</td><td>C2</td></tr><tr><td>Low</td><td>F</td><td>family</td><td>C2</td></tr><tr><td>Low</td><td>F</td><td>Sports</td><td>C1</td></tr><tr><td>Low</td><td>F</td><td>Sports</td><td>C2</td></tr><tr><td>high</td><td>M</td><td>family</td><td>C1</td></tr><tr><td>high</td><td>F</td><td>family</td><td>C2</td></tr><tr><td>high</td><td>F</td><td>family</td><td>C2</td></tr><tr><td>High</td><td>F</td><td>sports</td><td>C2</td></tr><tr><td>Low</td><td>M</td><td>sports</td><td>C2</td></tr><tr><td>Low</td><td>F</td><td>family</td><td>C2</td></tr><tr><td>low</td><td>M</td><td>sports</td><td>C1</td></tr></table>				Customer income	Gender	Car type	class	High	M	Family	C1	High	M	Sports	C1	High	M	family	C2	low	M	family	C2	Low	F	family	C2	Low	F	Sports	C1	Low	F	Sports	C2	high	M	family	C1	high	F	family	C2	high	F	family	C2	High	F	sports	C2	Low	M	sports	C2	Low	F	family	C2	low	M	sports	C1
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Que. No.	Question	Max. Marks
Q3	Solve any Two	20

i)	How does k-means clustering works? For the given dataset identify centroids in first iteration, with initial centroid as $C1 = (185, 72)$ and $C2 = (170, 56)$	10																					
<table border="1"> <thead> <tr> <th>Sample no</th><th>X</th><th>Y</th></tr> </thead> <tbody> <tr><td>1</td><td>185</td><td>72</td></tr> <tr><td>2</td><td>170</td><td>56</td></tr> <tr><td>3</td><td>168</td><td>60</td></tr> <tr><td>4</td><td>179</td><td>68</td></tr> <tr><td>5</td><td>182</td><td>72</td></tr> <tr><td>6</td><td>188</td><td>77</td></tr> </tbody> </table>			Sample no	X	Y	1	185	72	2	170	56	3	168	60	4	179	68	5	182	72	6	188	77
Sample no	X	Y																					
1	185	72																					
2	170	56																					
3	168	60																					
4	179	68																					
5	182	72																					
6	188	77																					
ii)	Why the cost function is called mean squared error function ? How do you arrive at the cost function for a single variable input?	10																					
iii)	How does ID3 algorithm works? Explain with suitable example.	10																					

Que. No.	Question	Max. Marks																																												
Q4	Solve any Two	20																																												
i)	What is the principle of Naïve Bays classifier. For the given dataset . Find the probability of brand X with respect to Blue Jeans? <table><tr><td>colour</td><td>Cloth type</td><td>Brand</td><td>On sale?</td></tr><tr><td>Blue</td><td>Jeans</td><td>Y</td><td>Yes</td></tr><tr><td>Blue</td><td>Shirt</td><td>X</td><td>No</td></tr><tr><td>Blue</td><td>Jeans</td><td>Y</td><td>Yes</td></tr><tr><td>Black</td><td>Jeans</td><td>Y</td><td>Yes</td></tr><tr><td>Black</td><td>Jeans</td><td>X</td><td>No</td></tr><tr><td>Black</td><td>Shirt</td><td>X</td><td>Yes</td></tr><tr><td>Black</td><td>Shirt</td><td>Y</td><td>Yes</td></tr><tr><td>Blue</td><td>Jeans</td><td>Y</td><td>No</td></tr><tr><td>Black</td><td>Jeans</td><td>X</td><td>Yes</td></tr><tr><td>Blue</td><td>shirt</td><td>X</td><td>Yes</td></tr></table>	colour	Cloth type	Brand	On sale?	Blue	Jeans	Y	Yes	Blue	Shirt	X	No	Blue	Jeans	Y	Yes	Black	Jeans	Y	Yes	Black	Jeans	X	No	Black	Shirt	X	Yes	Black	Shirt	Y	Yes	Blue	Jeans	Y	No	Black	Jeans	X	Yes	Blue	shirt	X	Yes	10
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ii)	How SVM works to identify hyperplane? How the margin is decided to separate hyperplanes?	10																																												
iii)	What is the need of covariance matrix in PCA? Find out the covariance matrix for the following three dimensional data? <table><tr><td>x</td><td>y</td><td>z</td></tr><tr><td>1</td><td>2</td><td>1</td></tr><tr><td>-1</td><td>1</td><td>3</td></tr><tr><td>4</td><td>3</td><td>-1</td></tr></table>	x	y	z	1	2	1	-1	1	3	4	3	-1	10																																
x	y	z																																												
1	2	1																																												
-1	1	3																																												
4	3	-1																																												

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	What is a slack variable in SVM ?	5
ii)	What is the need of dimensionality reduction?	5
iii)	State any two applications of logistic regression with justification	5
iv)	How confusion matrix is used to measure classifier accuracy?	5

v)	How regression is different than classification? Justify with example.	5
vi)	How bottom up approach is used in hierarchical clustering?	5



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1

25.05.2023 (E)

Semester: January 2023 –May 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01	Class: Third	Semester:VI (SVU 2020)
Programme: Computer Engineering B.Tech.		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116U01E627	Name of the Course: Micro Services and DevOPS	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What is DevOps? Explain the linkage of Agile and DevOps	5
ii)	Explain the concepts of Continuous Integration and Continuous Deployment along with tools and example	5
iii)	Explain Maven Build Life Cycle.	5
iv)	What are different phases of DevOps Life Cycle? Explain them in short.	5
v)	What are Spring Boot and Spring Cloud?	5
vi)	Explain the structure of Project Object Model.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	What is monolithic architecture? Enlist the Benefits and drawbacks of monolithic application	5
ii)	Compare and contrast Monolithic Vs Microservices architectures	5
	OR	
Q2 A		10
Q 2 B	Solve any One	10
i)	Explain the Step by step procedure for Developing the Spring Boot microservice using the CLI	10
ii)	What are Microservices? Explain their Principles and Characteristics in detail	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain the Step by step procedure for developing the Spring Boot Java microservice using STS	10
ii)	What is pipeline? What are different tools used to build a pipeline? Explain the fundamentals of declarative and scripted pipeline.	10
iii)	Explain the different development patterns, routing patterns, security patterns and deployment patterns in Core microservices	10

Q.2A	what is spring cloud ? why is spring cloud used ? what are the features of spring cloud?	10
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Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Explain in detail deployment of Microservices using Spring Cloud	10
ii)	Explain the different strategies to optimize the pipeline during implementation of effective CI/CD	10
iii)	Explain the relevance of different types of files used in Java microservices like POM file, Application.java, ApplicationTests.java etc.	10

Que. No.	Question	Max. Marks
Q5	Write a Short note on (any four)	20
i)	Version Control	5
ii)	Continuous Monitoring	5
iii)	Benefits and drawbacks of Microservices	5
iv)	Microservice logging and tracing patterns	5
v)	Spring Framework	5
vi)	RESTful microservices	5



25.05.2023 (E)

Semester: January 2023 –May 2023		Duration:3 Hrs.	
Maximum Marks: 100		Examination: ESE Examination	
Programme code: 01		Class: TY	Semester: VI (SVU 2020)
Programme: B. Tech in Computer Engineering			
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP	
Course Code: 116U01E628	Name of the Course: Applied Cryptography		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	<u>State</u> the relationship between Security Goals and Security attacks. Comment on which one the attack(s) harm which one the security goals.	5
ii)	Encrypt the message "Night is dark and full of terrors" using: a. Playfair cipher with key: Maharashtra b. Rail fence cipher with fence height=3	5
iii)	Compute: 1. $13 + 29$ in Z_{23} 2. Contents of the set Z_{11} 3. Contents of the set Z_{6^*}	1+2+2
iv)	What is cryptanalysis? Discuss cryptanalysis of substitution ciphers with an example.	5
v)	What are primitive roots? Comment on challenges and issues with primitive roots.	5
vi)	Give at least one application scenario for each of them: a. Session key b. PIN c. One Time Pad d. salt	1+1+1+2

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	What are initialization vectors? Comment on how inclusion and exclusion of initialization vectors affect security goals. Give a suitable example to support your claims.	5
ii)	Design the following, assume suitable data. a. A P-Box with 8 inputs - 8 outputs b. A P-Box with 8 inputs – 12 outputs c. A S-Box with 8 inputs – 4 outputs (8 inputs can be represented using binary code 000-111)	1+2+2
	OR	
Q2 A	Compare and contrast between stream ciphers and block ciphers on the basis of: a. Security : consider cryptography and cryptanalysis, both b. Speed and efficiency c. Ease of implementation in hardware and software d. Examples algorithms from both categories e. At least two application scenarios	10

Q 2 B	Solve any One	10
i)	Discuss significance and relationship of the following concepts in asymmetric key cryptography: a. Prime number generation b. Primality testing c. Factorization d. Exponentiation and e. Logarithm algorithms	10
ii)	Compare AES and DES based on security, Strength and weaknesses	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Discuss working of HMAC for Message authentication and integrity	10
ii)	Explain Diffie-Hellman Key exchange algorithm with an example. Discuss the man-in-middle attack possibility scenario too.	10
iii)	Discuss Digital signatures and their significance in the field of security.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Discuss in brief working of Kerberos. [Must include major steps from client authentication to using a service.]	10
ii)	Consider a scenario wherein you are required to give a program sourcecode to other party as a part of transaction. Discuss in brief various method(s) those could be used to achieve : a. Confidentiality b. Integrity c. Non-repudiation on senders side d. Non-repudiation on receiver's side	10
iii)	For the above scenario i.e. you are required to send a program sourcecode to other party as a part of transaction. Discuss in brief various method(s) those could be used to achieve : a. No modification/distribution/reverse engineering by the client party b. No interruption/interception while you are transmitting code online c. The client can prove their potential customers that they have certain code that works for a particular application	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	What are Discrete logarithms? Explain their significance in applied cryptography.	5
ii)	Discuss Secure Multiparty Computation with strengths, weaknesses and applications	5
iii)	Cryptographic Hash function requirements	5
iv)	Discuss Asymmetric encryption for Authentication	5
v)	Weakness(s) of RSA, State solutions over the same.	5
vi)	MD5: Strengths, weaknesses, applications	5



SOMAIYA
VIDYAVIHAR UNIVERSITY

25.05.2023 (E)

Semester: January 2023 –May 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01	Class: Third Year	Semester:VI (SVU 2020)
Programme: B.Tech (COMP.)		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116U01E629	Name of the Course: Cloud Computing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Define Cloud Computing. Explain its four deployment models.	5
ii)	Explain the concept of sandboxing in Google AppEngine. Give the examples of some of the operations not allowed in the sandbox.	5
iii)	State IaaS, PaaS and SaaS with respect to what they provide to the customer, product types associated with them and Vendors with their specific products providing them.	5
iv)	Explain how cloud computing technology is useful in productivity related applications giving examples.	5
v)	Explain the concept of containerization. What are its advantages over virtualization?	5
vi)	An eCommerce company needs to build its own IT infrastructure so as to facilitate scalability when needed and maximum uptime for its services. Further, it wants to provision the computing resources to all its departments like R&D, Analytics, Marketing etc. from the same infrastructure. It is desirable that the development work, experimentation going on in the different departments is properly isolated in their respective VMs and the problem in any VM doesn't affect the functioning of VMs of the other departments. Moreover, it would like the VMs to be independent of any host OS so as to make them isolated from its limitations. The company would like to buy heterogeneous types of processors so that it can cater to all kinds of computing needs. The company is also ready to recruit the experts for deployment and maintenance if all the above-mentioned requirements lead to complexity. Recommend the virtualization solution including mechanism, structure of virtualization for this company. Clearly justify your choice by stating how it addresses each specification given above.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Explain the concept of Federated clouds. Explain the needs in cloud computing that leads to the concept of Federated Cloud.	5
ii)	What is energy efficient computing? Why is it fundamental in Cloud Computing?	5
	OR	
Q2 A	Explain the Market Oriented Cloud Computing model with a neat diagram.	10
Q2 B	Solve any One	10
i)	Explain the following with respect to Amazon S3: a) Resource naming, b) Buckets, c) Objects and Metadata, d) Access control and security.	10
ii)	Explain the architecture of Google AppEngine with a neat diagram.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Compare Microsoft Hyper V and VMWare vSphere.	10
ii)	Explain the challenges in fully adopting cloud computing. Explain each one giving examples.	10
iii)	What issues/challenges does the logical and operational level of the Cloud Federation Stack address? Explain each one in detail.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	What are the third-party cloud services? Explain MetaCDN in detail.	10
ii)	Explain the high-level architecture for supporting energy-efficient resource allocation in Green Cloud Computing infrastructure with a neat diagram. Explain each component in detail.	10
iii)	Explain the Amazon EC2 instance life cycle.	10

Que. No.	Question	Max. Marks
Q5	Attempt any four of following :	20
i)	What do you understand by the term service with respect to cloud computing? What are the characteristics of a good service? Explain each characteristic in detail giving examples.	5
ii)	Define virtualization. Explain in brief the desktop, network, data and application virtualizations.	5
iii)	Explain the application services available in Google AppEngine.	5
iv)	Explain how cloud computing technology is useful in media applications giving examples.	5
v)	What is the vendor lock-in problem? What effect does it have on the commercial aspects of Cloud Computing?	5
vi)	Explain the concept of Virtual Market Place. What are its main features?	5



Semester: January 2023 –May 2023		Duration:3 Hrs.	
Maximum Marks: 100		Examination: ESE Examination	
Programme code:01		Class: TY	Semester: VI (SVU 2020)
Programme: B Tech Computer Engineering			
Name of the Constituent College:		Name of the department: Computer	
K. J. Somaiya College of Engineering			
Course Code: 116U01C602	Name of the Course: Information Security		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory			
3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Assume a hacker hacks into a network, copies a few files, defaces the Web page, and steals credit card numbers, how many different threat categories does this attack fall into?	5
ii)	List the different layers of an organization where security must be implemented to protect its operations?	5
iii)	How does HTTPS provide security in comparison with http protocol? Which all fields are get authenticated and encrypted in case of https protected messages?	5
iv)	What is the difference between link and end-to-end encryption?	5
v)	What's the difference between a legal issue and an ethical issue?	5
vi)	Is it possible to use the DES algorithm to generate message authentication code? Justify.	5

Que. No.	Question	Max. Marks
Q 2 B	Solve any One	10
i)	Given $p=19$, $q=23$, and $e=3$ Use RSA algorithm to find n , $\phi(n)$ and d .	10
ii)	Define buffer overflows attacks. Give an example. What are its security implications?	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	A news headline reads "The website of a company A is hacked and their homepage was replaced with an obscene message". What really happened and how can it be avoided?	10
ii)	Discuss the methodologies used for web application hacking?	10
iii)	Write the steps to read Email Headers and identify them as SPAM.	10

Q 2 A	Solve the following	05
i)	Compare and contrast AES and DES algorithm.	05
ii)	List best practices to stay safe from man-in-middle attack	
Q.2 A	Write short note on OWASP.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	What is a firewall? List the type of firewalls categorized by processing mode. Draw a schematic diagram of a packet filtering router used as a firewall and explain its function using a sample firewall rule.	10
ii)	How does PGP provide authentication and confidentiality for email services and for file transfer applications? Draw the block diagram and explain the components.	10
iii)	What are the important ethical issues in cybersecurity? Explain in detail.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	What requirements must a public key cryptosystem fulfill to a secured algorithm?	5
ii)	Differentiate between trojan horse and denial-of-service attacks.	5
iii)	How web security can be achieved? What are the different mechanisms?	5
iv)	If I'm on my laptop, here inside my company, and I have just plugged in my network cable. How many packets must leave my NIC in order to complete a traceroute to twitter.com?	5
v)	Explain laws and ethics in Information Security.	5
vi)	Explain how the DES algorithm can be strengthened by 3 DES. What is the effective key length of 3 DES? Justify.	5



SOMAIYA
VIDYAVIHAR UNIVERSITY

7 30/05/2023 (E)

Semester: Jan 2023 – May 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01	Class: TY	Semester: VI (SVU 2020)
Programme: B. Tech. in Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP
Course Code: 116U01C603	Name of the Course: Artificial Intelligence	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve Any Four	20
i)	Define and explain four approaches for AI.	5
ii)	Define the following term :- a) Agent (1) b) Rational agent (1) c) Omniscient Agent (1) d) Turing Test (2)	5
iii)	Write well-formed formulae for the following : a) Some students took French in spring 2001. (1.5) b) Every student who takes French passes it. (1.5) c) Only one student took Greek in spring 2001. (2)	5
iv)	Describe constraint satisfaction problems. While solving CSPs, which variable should be assigned next, and in what order should its values be tried?	5
v)	Differentiate between ADL and STRIPS language of planning.	5
vi)	Discuss the characteristics of expert system.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Consider the Water Jug Problem “You are given two jugs, a 4-gallon one and 3-gallon one. Neither has any measuring marker on it. A pump which has unlimited water which you can use to fill the jug, and ground on which water may be poured. How can you get exactly 2 gallons of water in the 4-gallon jug?” Describe the operator and production rules.	5
ii)	Discuss the properties of task environment for part picking robot.	5
OR		

Q2 A	Sketch the architecture of utility agent. Discuss the component with the help of suitable example.	10
Q2 B	<i>Solve Any One</i>	10
	Consider the graph given in Figure 1 below. Assume that initial state is S and goal state is I. Figure 1	
i)	Draw the state space tree (4). Find a path from the initial state to the goal state with cost using the following algorithm. - Breadth First Search (BFS - 3) - Depth First Search (DFS - 3)	10
ii)	Assuming the heuristic function : $h(S)=10$, $h(A) = 8$, $h(B)=5$, $h(C)=7$, $h(D)=3$, $h(E)=6$, $h(F)=5$, $h(G)=3$, $h(H)=1$, $h(I)=0$. Use the A* Search to find a path from the initial state to the goal state with cost.	10

Que. No.	Question	Max. Marks
Q3	<i>Solve the following.</i>	20
i)	Explain the knowledge engineering process for one bit full adder circuit.	10
ii)	Consider the following set of sentences. a. Whoever can read is literate. b. Dolphins are not literate. c. Some dolphins are intelligent. d. Prove/Goal - Some who is intelligent, cannot read. Convert the sentences in First Order Logic (4). Prove the sentence using resolution.(3+3)	10
OR		
ii)	Consider a new burglar alarm installed at home. It is fairly reliable at detecting a burglary, but also responds on occasion to minor earthquakes. There are two neighbours, John and Mary, who have promised to call the home owner at work	10

	when they hear the alarm. John always calls when he hears the alarm, but sometimes confuses the telephone ringing with the alarm and calls then, too. Mary, on the other hand, likes rather loud music and sometimes misses the alarm altogether. (Assume necessary data) - Draw topology of the network. - Associate a conditional probability table for each node.(2) - Given the evidence that neither a burglary nor an earthquake has occurred and both John and Mary call, Estimate the probability of ringing the alarm bell. (3) - What are the reasons that uncertainty gets introduced in problem solving?(3)	
Que. No.	Question	Max. Marks
Q4	Solve the following	20
i)	Explain the partial order planning algorithm for wearing shoes.	10
OR		
i)	Discuss the Steps to build an NLP? Give applications of NLP in various areas.	10
ii)	Discuss the alpha-beta pruning algorithm. Comment on its strength and weaknesses. Solve the given problem instance using alpha-beta pruning to compute final score and declare the winner. <pre> graph TD A((A)) --- B((B)) A --- C((C)) B --- D((D)) B --- E((E)) C --- F((F)) C --- G((G)) D --- D2[2] D --- D3[3] E --- E5[5] E --- E9[9] F --- F0[0] F --- F1[1] G --- G7[7] G --- G5[5] </pre>	4+2+4

Figure 2

Que. No.	Question	Max. Marks
Q5	Solve Any Four.	20
i)	Define the terms chromosome (1), fitness function (1), crossover (2) and mutation (1) used in Genetic Algorithm.	5
ii)	Discuss the problems in hill climbing algorithm solutions.	5
iii)	Discuss the heuristic property admissibility with example.	5
iv)	Discuss the Universal Instantiation and Existential Instantiation Elimination with example.	5
v)	Describe the architecture of the Expert System with a diagram.	5
vi)	Discuss the Learning process in Decision Tree.	5

7

23.5.2023 (E)



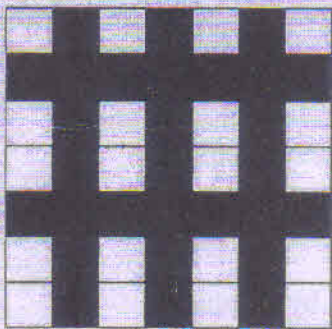
SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester:— January 2023 – April 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01	Class: TY	Semester: VI (SVU 2020)
Programme: B.Tech		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computers
Course Code: 116U01C601	Name of the Course: Digital Signal and Image Processing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks									
Q1	Solve any Four	20									
i)	Test the system for time invariance , where b is positive integer: $y(n) = x(n) - b x(n-1)$	5									
ii)	Given is a 3*3 image, plot its bit planes. <table border="1" data-bbox="548 925 933 1041"> <tr> <td>7</td><td>6</td><td>3</td></tr> <tr> <td>5</td><td>1</td><td>0</td></tr> <tr> <td>4</td><td>3</td><td>2</td></tr> </table>	7	6	3	5	1	0	4	3	2	5
7	6	3									
5	1	0									
4	3	2									
iii)	Explain Canny Edge Detector in short.	5									
iv)	Describe low pass frequency domain filter.	5									
v)	Distinguish between lossy and lossless compression (5 points).	5									
vi)	Determine whether the following signal is energy or power signal: $x(n) = \sin(\frac{\pi}{3}n)$	5									

Que. No.	Question	Max. Marks
Q2 A	Test the causality of the following systems:	10
i)	$y(n) = x(-n)$	05
ii)	$y(n) = x(3n)$	05
	OR	
Q2 A	Construct the block diagram and signal flow graph of the discrete time system whose input-output relations are described by following difference equation $y(n) = 0.2y(n-1) + 0.7x(n) + 0.9x(n-1)$	10
Q 2 B	Solve any One	10
i)	Perform auto correlation of the sequence, $x(n) = \{1, 2, 3, 4\}$ by graphical method.	10
ii)	Perform Circular Convolution of the two sequences $x_1(n)$ and $x_2(n)$, where (using graphical method only) $x_1(n) = \{0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6\}$ $\uparrow$ $x_2(n) = \{0.1, 0.3, 0.5, 0.7, 0.9, 1.1, 1.3, 1.5\}$ $\uparrow$	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	In an LTI system the input $x(n) = \{1, 2, 3\}$ and the impulse response $h(n) = \{-1, -1\}$. Determine the response of the LTI system by radix-2 DIT FFT.	10
ii)	Compute the discrete cosine transform (DCT) matrix for $N = 4$.	10
iii)	Perform KL transform for the following matrix: $X = \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Compute the Haar transform of the given digital image. $\begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 2 & 3 & 4 & 3 \\ 1 & 2 & 3 & 2 \end{bmatrix}$	10
ii)	Find Huffman Code for the following stream of data $\{a, a, a, a, a, a, b, b, b, c, c, c, c, c, d, d, d, d, d, d, d, d, e, e, e, e, f, f\}$	10
iii)	The input image and structuring elements are shown below. Find the hit or miss transformation for the input image and write all the steps in detail.    (a) Input image (b) Structuring element B (c) Structuring element $W-B$	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Dynamic Range Compression	5
ii)	Hough Transform	5
iii)	Median Filter	5
iv)	Vector Quantization	5
v)	Region Growing	5
vi)	Image Moments	5



Semester: January 2023 –May 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 54	Class: T Y	Semester:VI(SVU 2020)
Programme: Honours- Data Science and Analytics	B.Tech	
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116h54C601	Name of the Course: Advanced Data Mining	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What is descriptive data mining? Discuss about different descriptive data mining tasks.	5
ii)	What is Big Data? Discuss any 3 v's of Big data with example	5
iii)	Explain any 3 applications where data stream mining is useful?	5
iv)	Write the HITS algorithm	5
v)	Give any application of time series data mining. Also explain any 2 time series data mining task	5
vi)	Discuss with an example method to find the purity of clusters	5

Que. No.	Question	Max. Marks												
Q2 A	Solve the following	10												
i)	Discuss the features of Hadoop framework for processing Big Data	5												
ii)	Give example using k-means clustering a) Different initial centroid results in different clusters b) Sensitive to outliers	5												
	OR													
Q2 A	Explain Distributed Data Mining in detail.	10												
Q 2 B	Solve any One	10												
i)	A database has five transactions. Let min_sup_count be 2 and min_conf be 70%. <table border="1"><thead><tr><th>TID</th><th>Items bought</th></tr></thead><tbody><tr><td>T100</td><td>a,b,c</td></tr><tr><td>T200</td><td>b,c,d,e</td></tr><tr><td>T300</td><td>c,d</td></tr><tr><td>T400</td><td>a,b,d</td></tr><tr><td>T500</td><td>a,b,c</td></tr></tbody></table> Find all frequent patterns using Apriori algorithm . List any 3 valid association rules with support, confidence, lift	TID	Items bought	T100	a,b,c	T200	b,c,d,e	T300	c,d	T400	a,b,d	T500	a,b,c	10
TID	Items bought													
T100	a,b,c													
T200	b,c,d,e													
T300	c,d													
T400	a,b,d													
T500	a,b,c													
ii)	Illustrate Flajolet Martin algorithm with suitable example. Discuss an application of the algorithm.	10												

Que. No.	Question	Max. Marks																																				
Q3	Solve any Two	20																																				
i)	Term frequency matrix for 5 articles (A1 to A5) is shown below. Using Tf-idf score, find the similarity between articles? Identify the two articles that are most similar. <table><tr><th>Article/Terms</th><th>Trump</th><th>JNU</th><th>AAP</th><th>Corona</th><th>Divestiture</th></tr><tr><td>A1</td><td>14</td><td>1</td><td>0</td><td>6</td><td>3</td></tr><tr><td>A2</td><td>0</td><td>21</td><td>5</td><td>0</td><td>0</td></tr><tr><td>A3</td><td>0</td><td>15</td><td>18</td><td>0</td><td>5</td></tr><tr><td>A4</td><td>5</td><td>2</td><td>0</td><td>12</td><td>0</td></tr><tr><td>A5</td><td>0</td><td>0</td><td>5</td><td>0</td><td>10</td></tr></table>	Article/Terms	Trump	JNU	AAP	Corona	Divestiture	A1	14	1	0	6	3	A2	0	21	5	0	0	A3	0	15	18	0	5	A4	5	2	0	12	0	A5	0	0	5	0	10	10
Article/Terms	Trump	JNU	AAP	Corona	Divestiture																																	
A1	14	1	0	6	3																																	
A2	0	21	5	0	0																																	
A3	0	15	18	0	5																																	
A4	5	2	0	12	0																																	
A5	0	0	5	0	10																																	
ii)	Employ the DGIM algorithm. Shown below is a data stream with N=40 and current bucket configuration. End time size ↓ <table><tr><td>100</td><td>98</td><td>95</td><td>92</td><td>87</td><td>80</td><td>65</td></tr><tr><td>1</td><td>1</td><td>2</td><td>2</td><td>4</td><td>8</td><td>0</td></tr></table> Suppose that at times 101 through 105, 1's appear in the stream. Compute the set of buckets that would exist in the system at time 105. Also compute the number of 1's in latest k=12 bits of the window.	100	98	95	92	87	80	65	1	1	2	2	4	8	0	10																						
100	98	95	92	87	80	65																																
1	1	2	2	4	8	0																																
iii)	Apply Girvan-Newman algorithm and calculate the betweenness centrality for edges on social graph given below	10																																				

Que. No.	Question	Max. Marks																																				
Q4	Solve any Two	20																																				
i)	Discuss 4 components of Apache Hadoop framework	10																																				
ii)	What does apriori property state? How is the use of apriori property in apriori algorithm of finding frequent patterns?	10																																				
iii)	Consider the following distance matrix given below. Apply agglomerative clustering using single link and complete link approach to find hierarchy of clustering. Clearly show the steps of construction of dendrogram	10																																				
<table><tr><th>Item</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr><tr><td>A</td><td>0</td><td>1</td><td>2</td><td>2</td><td>3</td></tr><tr><td>B</td><td>1</td><td>0</td><td>2</td><td>4</td><td>3</td></tr><tr><td>C</td><td>2</td><td>2</td><td>0</td><td>1</td><td>5</td></tr><tr><td>D</td><td>2</td><td>4</td><td>1</td><td>0</td><td>3</td></tr><tr><td>E</td><td>3</td><td>3</td><td>5</td><td>3</td><td>0</td></tr></table>			Item	A	B	C	D	E	A	0	1	2	2	3	B	1	0	2	4	3	C	2	2	0	1	5	D	2	4	1	0	3	E	3	3	5	3	0
Item	A	B	C	D	E																																	
A	0	1	2	2	3																																	
B	1	0	2	4	3																																	
C	2	2	0	1	5																																	
D	2	4	1	0	3																																	
E	3	3	5	3	0																																	

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Discuss 5 steps of KDD process	5
ii)	Write short notes naïve bayes text classification approach	5
iii)	Consider the Web graph with three nodes 1, 2, 3. The links are as follows: 1->2, 2->1, 2->3. Compute the page rank with $\beta=0.5$	5
iv)	Write the algorithm for sequence pattern discovery.	5
v)	Find the Jaccard coefficient (J_c) for the query q and docs d1 and d2 below. Query: top university (set q) Doc 1: university of California (set d1) Doc 2: best university in USA (set d2) What are the limitations of Jaccard score ?	5
vi)	Write a short notes on architecture of Distributed data mining system	5

01.06.2023(E)

Maximum Marks: 100	Semester: January 2023 – May 2023	Duration: 3 Hrs.
Programme code: 55	Examination: ESE Examination	
Programme: Honours in Cyber Security & Forensics	Class: TY	Semester: VI (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering	
Course Code: 116h55C601	Name of the Course: Vulnerability Assessment & Penetration Testing	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Differentiate between Vulnerability Assessment and Penetration Testing (5 Points each)	5
ii)	Differentiate in detail between Red Team and Blue Team (5 Points each)	5
iii)	What is DNS enumeration?? How is DNS zone transfer enumeration done using nslookup?	5
iv)	Explain OS fingerprinting in detail.	5
v)	Explain in brief about all Modules of Metasploit framework.	5
vi)	What results you will get from Port Scanning? Explain.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	List OWASP top 10 Vulnerabilities in 2021.	5
ii)	How is footprinting done through social engineering? Explain.	5
	OR	
Q2 A	What is network traffic scanning? Explain any tool used for the same.	10
Q2 B	Solve any One	10
i)	Explain with help of neat diagram any 4 different types of Port Scanning	10
ii)	What is Enumeration? Explain in detail SMB enumeration and countermeasures against it.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain SSH Tunnelling? Explain the scenario in brief where SSH tunnelling is required.	10
ii)	What is ARP spoofing? Write a detailed note on detection and prevention of ARP spoofing.	10
iii)	Define session hijacking. Describe the steps involved in session hijacking.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Explain System Hacking steps in detail.	10
ii)	Explain Windows Privilege Escalation Techniques in detail.	10
iii)	Explain Fuzzing and its types with the help of an example.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Pivoting	5
ii)	SSL Stripping attack	5
iii)	Bind Shell and Reverse Shell	5
iv)	Incident Response Life Cycle	5
v)	Buffer Overflow vulnerabilities	5
vi)	CVE and CVSS	5